

SIT327 HD1 — Network Forensics Automation Project Report

Traditional Packet Inspection versus AI-Assisted Automated Network Forensics Triage: A Comparative Study Using the UNSW-NB15 Dataset

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Unit: SIT327

Assessment: HD1 Project Report

Project Title: AI-Assisted Network Forensics Automation

Introduction

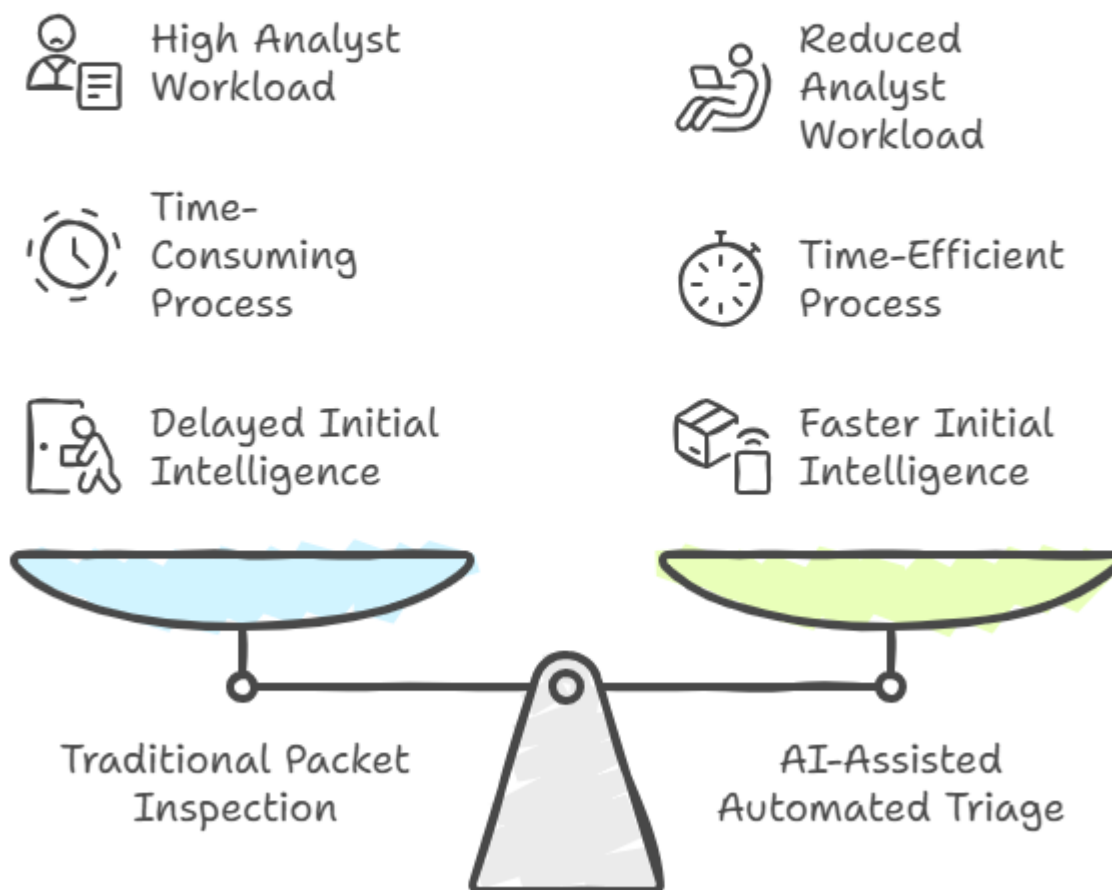
Network forensics plays a critical role in modern cybersecurity investigations by enabling analysts to inspect suspicious network behaviour, reconstruct attack activity, and identify malicious communication patterns. Traditional packet inspection generally relies on the manual examination of captured traffic using tools such as Wireshark, Tshark, or exported traffic summaries. Although this method provides detailed visibility, it often requires repetitive filtering, counting, note taking, and manual report preparation, particularly when dealing with larger traffic volumes. This creates significant analyst workload and can delay the production of initial forensic intelligence.

With the increasing availability of artificial intelligence (AI) and local large language models (LLMs), there is growing interest in whether some parts of the initial network forensic triage process can be automated. Instead of requiring the investigator to manually inspect every suspicious record and prepare written findings, an automated system may be capable of extracting important traffic indicators, summarising suspicious behaviour, and generating a preliminary forensic report.

This project investigates the practical difference between these two approaches by comparing traditional manual packet inspection against an AI-assisted automated network forensics triage workflow using suspicious traffic records extracted from the UNSW-NB15 intrusion detection dataset. Recent cyberattacks and large-scale security breaches have demonstrated the growing operational and financial impact of delayed incident detection and investigation. According to IBM's Cost of a Data Breach Report 2024, the global average cost of a data breach continues to increase each year, with delayed forensic investigation and response significantly increasing organisational losses. Security Operations Centres (SOCs) are increasingly challenged by large volumes of network telemetry and security logs, creating substantial analyst workload during forensic triage and incident response processes. As a result, there is growing interest in AI-assisted

security analysis systems capable of accelerating early-stage forensic investigation and reducing repetitive analyst tasks.[4]

AI-Assisted Triage Enhances Network Forensics Efficiency



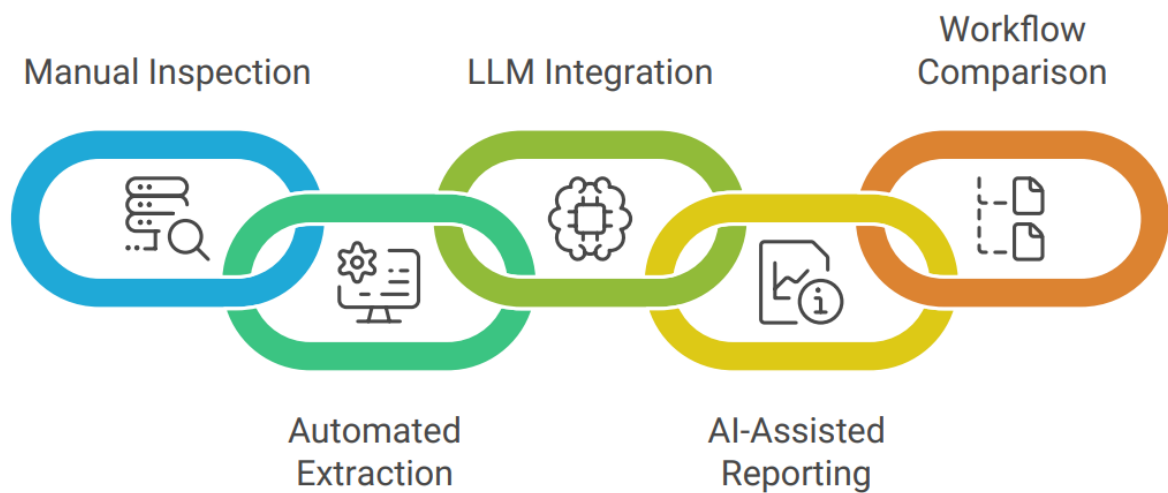
Research Objectives

The main objective of this study is to evaluate whether AI-assisted automation can reduce the repetitive workload involved in the early stages of network forensic analysis while still producing useful initial forensic observations.

The project specifically aims to:

- manually inspect a selected malicious traffic sample using traditional forensic observation methods;
- develop a Python-based automated traffic evidence extractor;
- integrate the extracted evidence with a locally hosted LLM using Ollama;
- generate an AI-assisted forensic triage report; and
- compare both workflows in terms of analyst effort, reporting speed, consistency, and scalability.

AI-Assisted Network Forensics



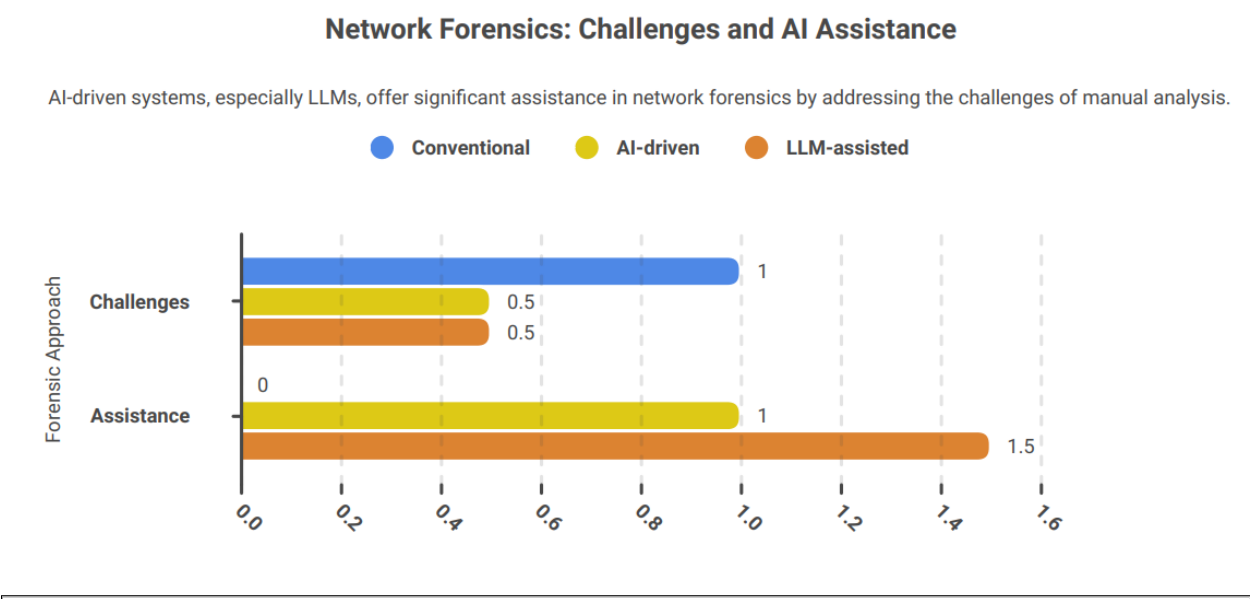
Literature Review

Recent research has shown that conventional network forensics faces increasing challenges due to the growing volume and complexity of modern network traffic, which makes manual evidence review slower and more demanding for investigators. Rizvi et al. explain that AI-driven forensic systems can significantly assist investigators by reducing repetitive manual analysis and generating machine-supported forensic intelligence [1]. Their study highlights that automated interpretation can support analysts during the early evidence triage phase where time pressure is high.

The UNSW-NB15 dataset has also become a widely accepted benchmark for evaluating modern intrusion detection and traffic analysis approaches because it contains labelled malicious and benign traffic records across multiple attack categories. More et al. identify UNSW-NB15 as a strong dataset for feature-based cyber analysis due to its realistic mixture of protocol behaviour, attack labels, and connection characteristics [2].

More recent studies involving large language models have suggested that LLMs can be used not only for text generation but also for cybersecurity reasoning when structured network evidence is provided. Emerging research into LLM-assisted intrusion interpretation indicates that converting network features into concise evidence summaries allows AI models to produce understandable analyst-style findings and recommended actions [3].

These studies collectively suggest that combining deterministic Python feature extraction with AI reasoning may offer a practical enhancement to traditional network forensic triage.



Methodology

This project used a comparative experimental methodology consisting of two separate forensic workflows applied to the same suspicious traffic sample so that both approaches could be observed under identical conditions.

The original UNSW-NB15 training dataset was first processed using a Python export script. Malicious traffic records were filtered by selecting all rows where the label field indicated attack activity. Relevant forensic columns including protocol information, connection states, packet statistics, byte counts, and attack categories were extracted and exported into a readable CSV file.

A total of 1000 malicious traffic records were selected for analysis. The resulting suspicious traffic sample contained several dominant attack categories including Fuzzers, Backdoor, Analysis, Reconnaissance, and Shellcode.

Stage 1 — Dataset Preparation

The preprocessing workflow converted the original Parquet dataset into a structured CSV format to simplify evidence review and ensure that both investigative workflows used the exact same evidence source.

Workflow:

UNSW-NB15 Dataset → Python Export Script → Suspicious Traffic CSV

Generated file:

`unsw_attack_sample.csv`

Stage 2 — Traditional Manual Inspection

Under the traditional workflow, the suspicious CSV traffic sample was manually inspected by reviewing:

- attack category distribution;
- protocol diversity;
- service behaviour;
- connection state behaviour; and
- packet and byte statistics.

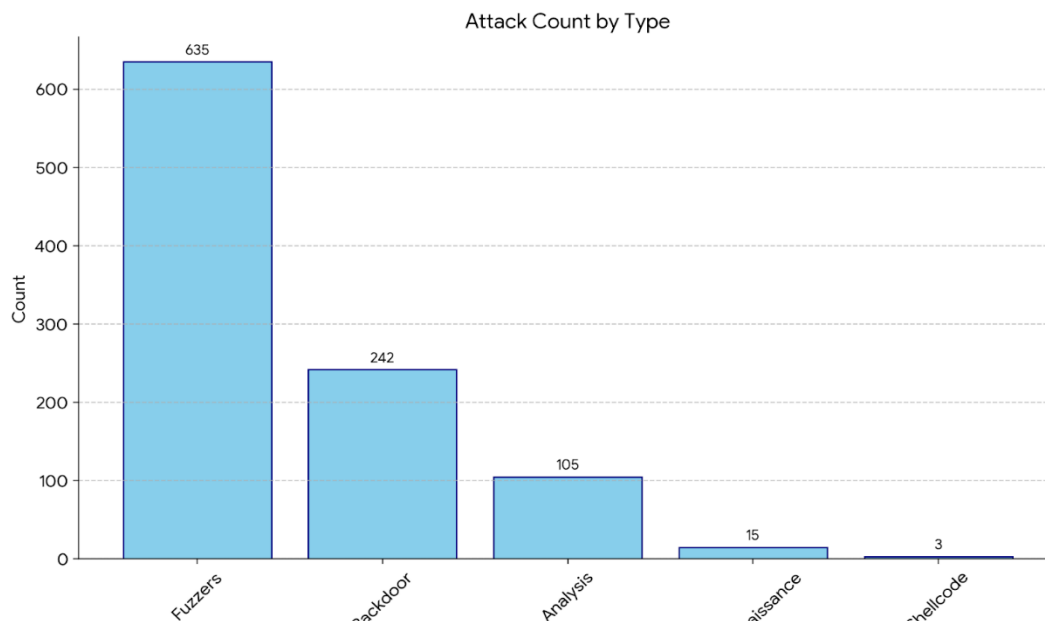
Manual observations were then written into a baseline forensic summary.

Manual Findings

The selected sample contained 1000 malicious traffic records extracted from the UNSW-NB15 dataset.

Attack category distribution:

- **Fuzzers: 635 records**
- **Backdoor: 242 records**
- **Analysis: 105 records**
- **Reconnaissance: 15 records**
- **Shellcode: 3 records**



Distribution of malicious attack categories extracted from the UNSW-NB15 suspicious traffic sample.

The dominant category within the sample was Fuzzers activity. This type of traffic is commonly associated with vulnerability discovery attempts where attackers send malformed or unexpected inputs to test system behaviour and identify weaknesses.

A significant number of Backdoor records were also identified, which may indicate attempts to establish unauthorized remote access or persistence mechanisms within a target environment. The Analysis category suggests probing or investigative behaviour against systems before exploitation attempts occur.

The protocol distribution showed that TCP traffic represented the largest proportion of suspicious records, followed by UDP and several uncommon protocol types. The presence of a wide range of unusual protocols may indicate attempts to evade standard monitoring or generate abnormal traffic patterns.

Most records were associated with unidentified services, while a smaller number involved FTP and HTTP services. This suggests that much of the malicious traffic did not map cleanly to common application-layer services.

The connection state distribution showed a high number of INT states compared with FIN states. This indicates that many connections were interrupted or not completed normally, which may reflect unstable or intentionally abnormal communication attempts during malicious activity.

Average packet and byte statistics also showed higher source-side activity compared with destination responses, which is consistent with aggressive outbound probing or attack-oriented traffic behaviour.

Generated file:

manual_inspection_report.md

Stage 3 — AI-Assisted Automated Analysis

Under the automated workflow, a Python forensic analyzer was developed to ingest the same suspicious traffic sample and automatically calculate traffic statistics including:

- attack category frequency;
- protocol distribution;
- service distribution;
- connection state frequency; and
- average packet and byte values.

The extracted indicators were structured into a forensic evidence block and submitted to a locally hosted Mistral 7B model through Ollama using Python subprocess communication.

The LLM was prompted to act as a professional network forensic investigator and generate:

- suspicious activity summary;
- likely attack behaviour;
- severity assessment; and
- recommended investigative actions.

The generated forensic analysis was then automatically saved as a structured report.

Generated file:

forensic_report.txt

Stage 4 — Comparative Evaluation

Both workflows were compared using the same suspicious traffic sample in order to evaluate:

- analyst workload;
- evidence extraction speed;
- reporting consistency; and
- scalability for larger datasets.

Comparative Findings

Evaluation Metric	Traditional Manual Workflow	AI-Assisted Workflow
Evidence counting	Manual	Automatic

Evaluation Metric	Traditional Manual Workflow	AI-Assisted Workflow
Traffic summarisation	Manual	Automatic
Report generation	Manual writing required	AI generated
Analyst workload	High	Reduced
Processing speed	Slower	Faster
Scalability for larger datasets	Limited	Strong
Investigative consistency	Analyst dependent	Standardised

The comparative evaluation showed that AI-assisted automation significantly reduced repetitive analyst workload during the initial forensic triage stage while also improving processing speed and investigative consistency. However, the results also demonstrated that human analyst validation remains essential when interpreting complex or high-risk security incidents.

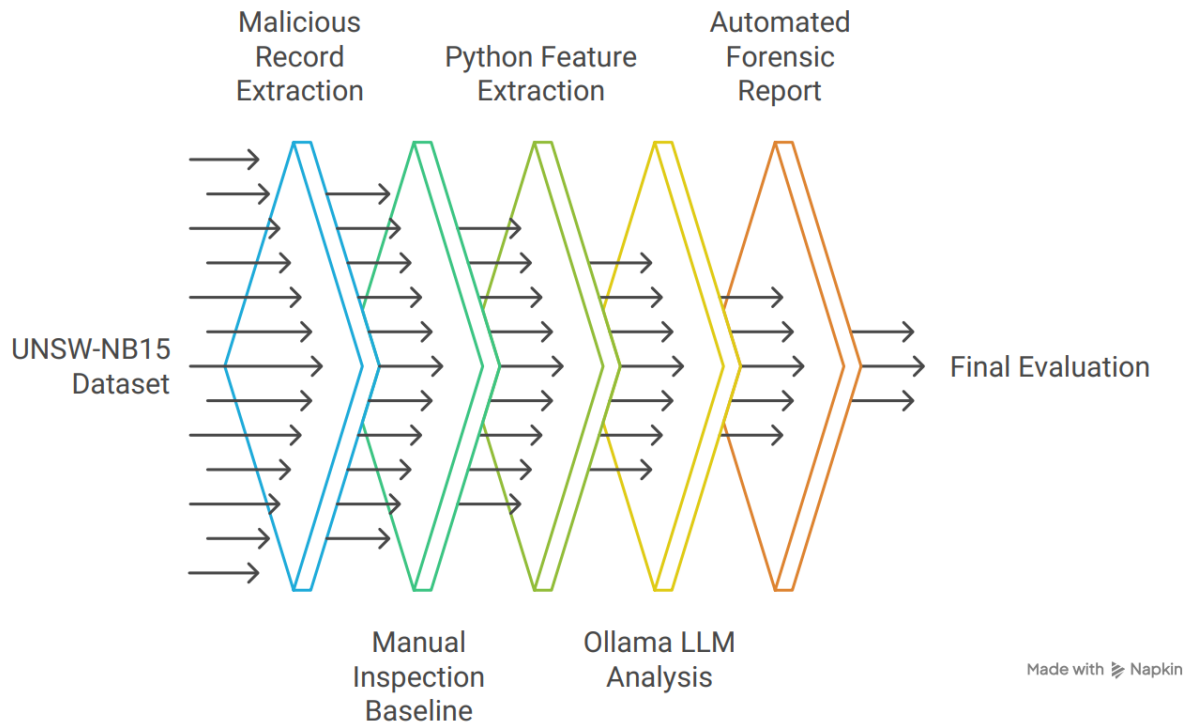
The traditional workflow provided useful human interpretation but required significant repetitive effort during evidence review and report preparation.

The AI-assisted workflow successfully automated the extraction and summarisation process while still producing meaningful forensic observations regarding fuzzing behaviour, backdoor activity, reconnaissance indicators, and suspicious protocol diversity.

The experiment demonstrated that AI-assisted automation can substantially reduce repetitive forensic triage workload while accelerating the generation of initial investigative intelligence.

However, the project also showed that human validation remains important because AI-generated analysis may still require analyst review for accuracy and contextual interpretation.

Data Analysis Process Funnel



Outcomes

The manual inspection workflow successfully identified that the traffic sample was dominated by Fuzzers activity, with significant proportions of Backdoor and Analysis attacks. However, this method required repeated counting, visual inspection, note taking, and manual narrative preparation.

The AI-assisted automated workflow produced the same traffic evidence distributions automatically and generated a structured forensic interpretation within seconds after execution of the Python analyzer. The automated report identified suspicious fuzzing behaviour, interpreted mixed attack activity as possible intrusion attempts, assigned a high severity level, and proposed relevant investigative actions such as host isolation, enhanced monitoring, and suspicious traffic investigation.

The comparative findings showed that while traditional packet inspection remains valuable for detailed human verification, AI-assisted automated triage substantially reduced analyst workload, accelerated initial reporting, and provided a standardised first-response forensic summary. The automated workflow also demonstrated stronger scalability when processing larger suspicious datasets.

Limitations of AI-Assisted Analysis

Although the AI-assisted workflow demonstrated strong capability for accelerating forensic triage and generating structured investigative summaries, several limitations remain. The quality of AI-generated findings depends heavily on the quality and structure of the provided evidence. Large language models may occasionally produce incorrect interpretations, overlook subtle indicators, or generate false positives if the input data is incomplete or noisy.

Additionally, the AI model used in this project does not possess real-time situational awareness and should not be considered a replacement for experienced forensic investigators. Human validation remains essential, particularly during critical investigations involving legal evidence, incident attribution, or high-risk security decisions.

The project also relied on a locally hosted language model with limited parameter size, which may affect analytical depth compared with larger enterprise-scale AI systems.

Conclusion

This project demonstrated the practical difference between traditional manual network forensic inspection and AI-assisted automated forensic triage using the UNSW-NB15 intrusion detection dataset.

The experiment showed that while manual inspection remains valuable for detailed human validation and investigative reasoning, AI-assisted automation significantly reduces repetitive workload and accelerates the generation of initial forensic intelligence.

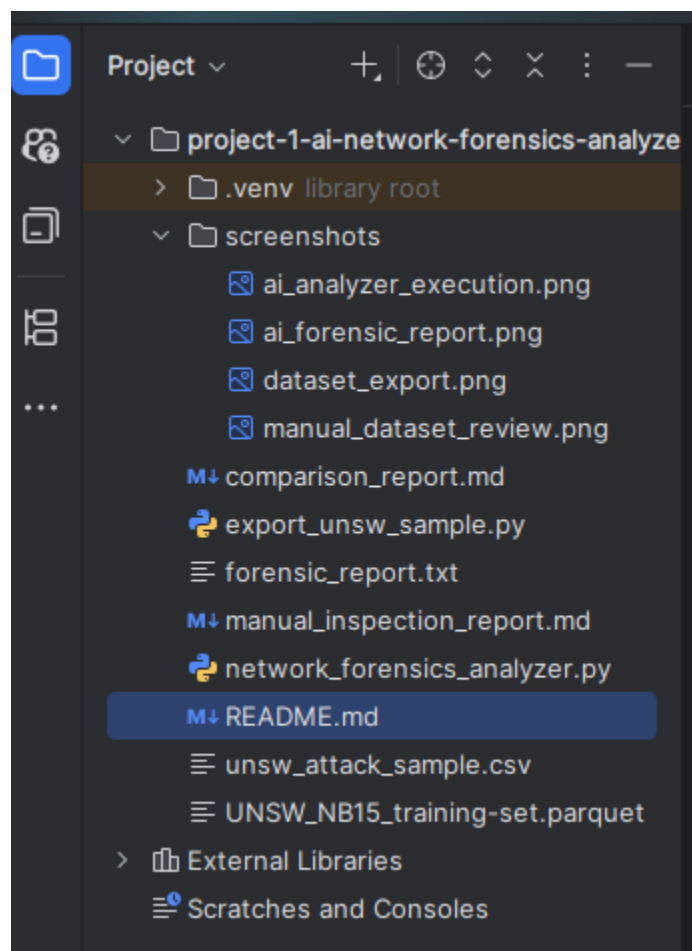
The combination of Python-based evidence extraction and locally hosted LLM reasoning successfully produced structured forensic summaries and investigative recommendations from suspicious traffic data. The project also highlighted the growing potential of AI-assisted cybersecurity workflows for supporting future SOC and incident response operations.

References

- [1] S. Rizvi, M. Scanlon, J. Sheppard and P. Reinecke, “Artificial Intelligence in Network Forensics: Investigative Opportunities and Challenges,” 2022.
 - [2] S. More, P. Khandait, V. Kharat and V. Mhaske, “Enhanced Intrusion Detection Systems Performance with UNSW-NB15 Dataset Using Machine Learning,” 2024.
 - [3] “Large Language Model Assisted Intrusion Reasoning Using Structured UNSW-NB15 Traffic Evidence,” arXiv preprint, 2025.
 - [4] IBM, “Cost of a Data Breach Report 2024,” IBM Security, 2024.
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Appendix — Technical Implementation Evidence

Figure A1 — Final Project Structure



Caption:

Figure A1: Final project implementation structure in PyCharm.

Listing A1 — Dataset Export Script

```
attack_df = df[df["label"] == 1]

attack_sample = attack_df[selected_columns].head(1000)

attack_sample.to_csv("unsw_attack_sample.csv", index=False)
```

Caption:

Listing A1: Python preprocessing script used to extract suspicious traffic records from the UNSW-NB15 dataset.

Listing A2 — Ollama Integration

```
result = subprocess.run(
    ["ollama", "run", "mistral:7b"],
    input=prompt,
    text=True,
    capture_output=True,
    encoding="utf-8"
)
```

Caption:

Listing A2: Python subprocess communication with the local Mistral 7B model through Ollama.

Figure A2 — AI Analyzer Execution

```
1 import pandas as pd
2 import subprocess
3 from datetime import datetime
4 import re
5
6 print("Loading UNSW attack traffic sample...")
7
8 df = pd.read_csv("unsw_attack_sample.csv")
9
10 # Basic forensic statistics
11 total_records = len(df)
12 attack_distribution = df["attack_cat"].value_counts()
13 protocol_distribution = df["proto"].value_counts()
14 service_distribution = df["service"].value_counts()
15 state_distribution = df["state"].value_counts()
16
17 avg_spkts = round(df["spkts"].mean(), 2)
18 avg_dpks = round(df["dpkts"].mean(), 2)
19 avg_sbytes = round(df["sbytes"].mean(), 2)
20 avg_dbytes = round(df["dbytes"].mean(), 2)
21
22 # Build evidence summary for AI
23 evidence = f"""
24 Total suspicious records analysed: {total_records}
25
Run network_forensics_analyzer x
Loading UNSW attack traffic sample...
Traffic evidence extracted.
Contacting local AI forensic analyst...
AI forensic report generated successfully.
Saved as forensic_report.txt
Process finished with exit code 0
```

ai_analyzer_execution.png

Caption:

Figure A2: Execution of the AI-assisted network forensic analyzer.

Figure A3 — Generated AI Forensic Report

=== AI FORENSIC ANALYSIS ===

1. Summary of suspicious findings: The network traffic analysis shows a significant presence of Fuzzers (635 instances) and Backdoors (242 instances), which are known attack categories in the UNSW-NB15 intrusion dataset. Other suspicious activities include Reconnaissance, Shellcode, Analysis, and various protocols that are not commonly used, such as kryptol, kryptolan, mfe-nsp, pvp, wb-expak, and br-sat-mon. The service distribution shows a large number of connections to an unidentified service (-937 instances), with a smaller number of connections to FTP (38 instances) and HTTP (22 instances).

2. Likely attack behaviour present: The high occurrence of Fuzzers indicates a potential Denial-of-Service (DoS) or reconnaissance attack, as Fuzzers are used to send malformed packets to a system in order to crash it or gather information about its vulnerabilities. Backdoors suggest the presence of persistent remote access, allowing an attacker to control compromised systems undetected. Reconnaissance activities may be employed before launching more targeted attacks.

3. Severity assessment: Given the large number of Fuzzers and Backdoor instances, the severity level of this network traffic is high. The attack could potentially lead to a successful intrusion, data theft, or system crashes causing service disruptions.

4. Why this traffic would require forensic attention: Analyzing this traffic is essential to determine if any systems have been compromised, identify the attacker's tactics, and prevent further attacks. The presence of Fuzzers, Backdoors, and uncommon protocols could indicate that an advanced persistent threat (APT) is active on the network.

5. Recommended investigative actions:

- Identify systems affected by the Fuzzers and Backdoor traffic to determine the extent of the compromise.
- Review log files from affected systems for indicators of compromise (IoCs), such as malicious IP addresses, unusual network activity patterns, or modified system configurations.
- Investigate uncommon protocols and services to identify potential command-and-control (C2) channels or data exfiltration routes.
- Update security controls and firewall rules to block known attack vectors and prevent further intrusion attempts.
- Implement network monitoring tools to detect similar traffic patterns in the future.
- Conduct post-incident response actions, such as patching identified vulnerabilities, cleaning affected systems, and notifying affected parties.

ai_forensic_report.png

Caption:

Figure A3: Generated AI forensic triage report.

Figure A4 — Dataset Export and Attack Distribution

```
1 import pandas as pd
2
3 df = pd.read_parquet("UNSW_NB15_training-set.parquet")
4
5 # label = 1 means attack traffic
6 attack_df = df[df["label"] == 1]
7
8 selected_columns = [
9     "dur", "proto", "service", "state",
10     "spkts", "dpkts", "sbytes", "dbytes",
11     "rate", "sload", "dload",
12     "ct_src_dport_ltm", "ct_dst_sport_ltm",
13     "is_ftp_login", "ct_ftp_cmd", "ct_flw_http_mthd",
14     "attack_cat", "label"
15 ]
16
17 attack_sample = attack_df[selected_columns].head(1000)
18
19 attack_sample.to_csv("unsw_attack_sample.csv", index=False)
20
21 print("Export complete. 1000 attack traffic records saved as unsw_attack_sample.csv")
22 print("Attack categories in sample:")
23 print(attack_sample["attack_cat"].value_counts())
```

Run export_unsw_sample x

↑ ↓ ⇄ ⇅ ⏏

```
"C:\Users\asim tanweer\PycharmProjects\project-1-ai-network-forensics-analyzer\.venv\Scr
Export complete. 1000 attack traffic records saved as unsw_attack_sample.csv
Attack categories in sample:
attack_cat
Fuzzers          635
Backdoor         242
Analysis         105
Reconnaissance   15
Shellcode         3
DoS              0
Exploits         0
Generic          0
Normal          0
Worms            0
Name: count, dtype: int64

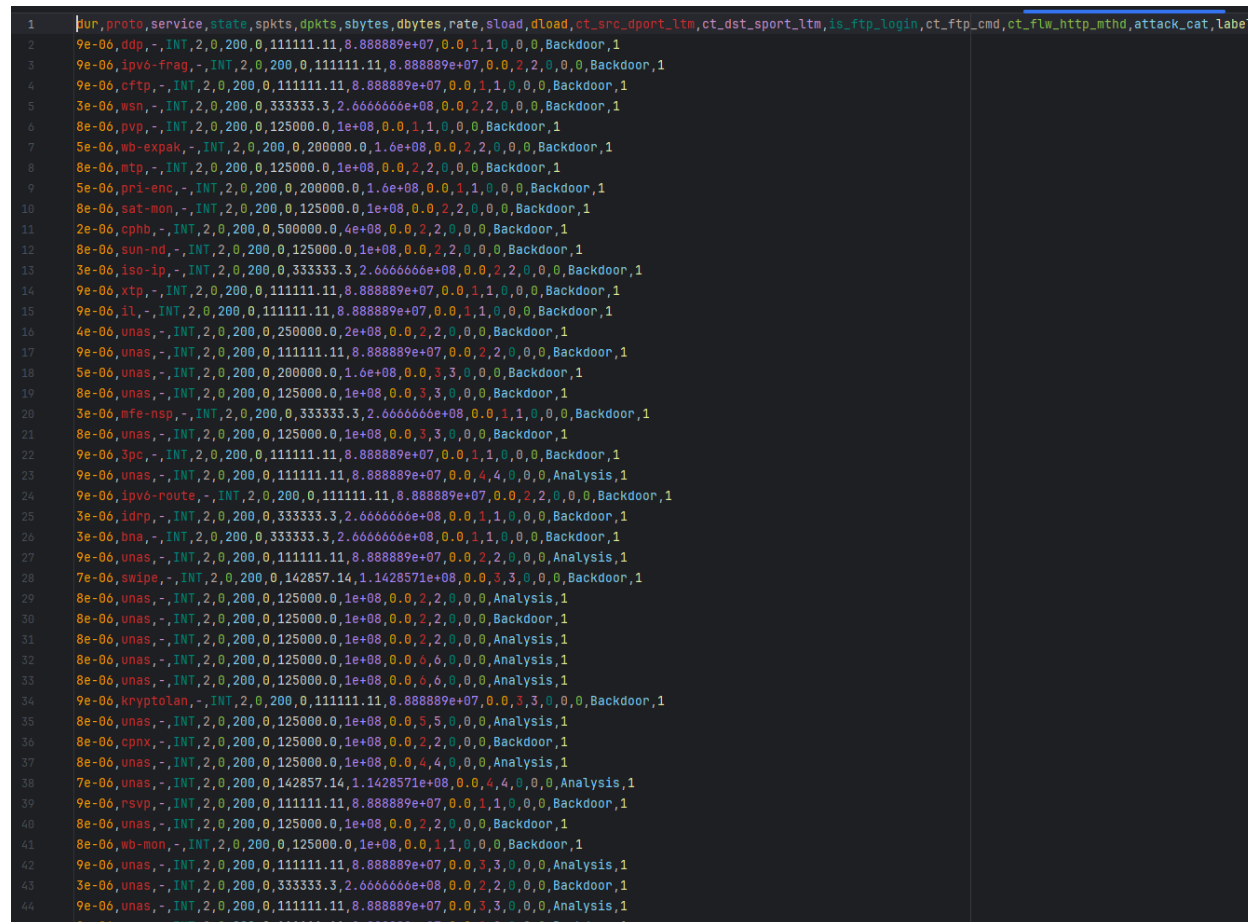
Process finished with exit code 0
|
```

dataset_export.png

Caption:

Figure A4: Export of malicious traffic records and attack category distribution.

Figure A5 — Manual Dataset Review



	lur, proto, service, state, spkts, dpkts, sbytes, dbytes, rate, sload, dload, ct_src_dport_ltm, ct_dst_sport_ltm, is_ftp_login, ct_ftp_cmd, ct_flw_http_mthd, attack_cat, label
1	9e-06, ddp, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 1, 1, 0, 0, 0, Backdoor, 1
2	9e-06, ipv6-frag, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
3	9e-06, cftp, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 1, 1, 0, 0, 0, Backdoor, 1
4	3e-06, wsn, -, INT, 2, 0, 200, 0, 333333.3, 2.666666e+08, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
5	8e-06, pvp, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 1, 1, 0, 0, 0, Backdoor, 1
6	5e-06, wb-expak, -, INT, 2, 0, 200, 0, 200000.0, 1.6e+08, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
7	8e-06, mtp, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
8	5e-06, pri-enc, -, INT, 2, 0, 200, 0, 200000.0, 1.6e+08, 0.0, 1, 1, 0, 0, 0, Backdoor, 1
9	8e-06, sat-mon, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
10	2e-06, cphb, -, INT, 2, 0, 200, 0, 500000.0, 4e+08, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
11	8e-06, sun-nd, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
12	3e-06, iso-ip, -, INT, 2, 0, 200, 0, 333333.3, 2.666666e+08, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
13	9e-06, xtp, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 1, 1, 0, 0, 0, Backdoor, 1
14	9e-06, il, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 1, 1, 0, 0, 0, Backdoor, 1
15	4e-06, unas, -, INT, 2, 0, 200, 0, 250000.0, 2e+08, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
16	9e-06, unas, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
17	5e-06, unas, -, INT, 2, 0, 200, 0, 200000.0, 1.6e+08, 0.0, 3, 3, 0, 0, 0, Backdoor, 1
18	8e-06, unas, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 3, 3, 0, 0, 0, Backdoor, 1
19	3e-06, mfe-nsp, -, INT, 2, 0, 200, 0, 333333.3, 2.666666e+08, 0.0, 1, 1, 0, 0, 0, Backdoor, 1
20	8e-06, unas, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 3, 3, 0, 0, 0, Backdoor, 1
21	9e-06, spc, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 1, 1, 0, 0, 0, Backdoor, 1
22	9e-06, unas, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 4, 4, 0, 0, 0, Analysis, 1
23	9e-06, ipv6-route, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
24	3e-06, idrp, -, INT, 2, 0, 200, 0, 333333.3, 2.666666e+08, 0.0, 1, 1, 0, 0, 0, Backdoor, 1
25	3e-06, bna, -, INT, 2, 0, 200, 0, 333333.3, 2.666666e+08, 0.0, 1, 1, 0, 0, 0, Backdoor, 1
26	9e-06, unas, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 2, 2, 0, 0, 0, Analysis, 1
27	7e-06, swipe, -, INT, 2, 0, 200, 0, 142857.14, 1.428571e+08, 0.0, 3, 3, 0, 0, 0, Backdoor, 1
28	8e-06, unas, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 2, 2, 0, 0, 0, Analysis, 1
29	8e-06, unas, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
30	8e-06, unas, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 2, 2, 0, 0, 0, Analysis, 1
31	8e-06, unas, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 2, 2, 0, 0, 0, Analysis, 1
32	8e-06, unas, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 6, 6, 0, 0, 0, Analysis, 1
33	8e-06, unas, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 6, 6, 0, 0, 0, Analysis, 1
34	9e-06, kryptolan, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 3, 3, 0, 0, 0, Backdoor, 1
35	8e-06, unas, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 5, 5, 0, 0, 0, Analysis, 1
36	8e-06, cpnx, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
37	8e-06, unas, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 4, 4, 0, 0, 0, Analysis, 1
38	7e-06, unas, -, INT, 2, 0, 200, 0, 142857.14, 1.428571e+08, 0.0, 4, 4, 0, 0, 0, Analysis, 1
39	9e-06, rsvp, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 1, 1, 0, 0, 0, Backdoor, 1
40	8e-06, unas, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
41	8e-06, wb-mon, -, INT, 2, 0, 200, 0, 125000.0, 1e+08, 0.0, 1, 1, 0, 0, 0, Backdoor, 1
42	9e-06, unas, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 3, 3, 0, 0, 0, Analysis, 1
43	3e-06, unas, -, INT, 2, 0, 200, 0, 333333.3, 2.666666e+08, 0.0, 2, 2, 0, 0, 0, Backdoor, 1
44	9e-06, unas, -, INT, 2, 0, 200, 0, 111111.11, 8.888889e+07, 0.0, 3, 3, 0, 0, 0, Analysis, 1

manual_dataset_review.png

Caption:

Figure A5: Manual inspection of suspicious traffic records extracted from the UNSW-NB15 dataset.

Appendix B — Manual Inspection Report Excerpt

The following excerpt shows the manually prepared forensic observations generated from the suspicious UNSW-NB15 traffic sample.

Manual Findings

The selected sample contained 1000 malicious traffic records extracted from the UNSW-NB15 dataset.

Attack category distribution:

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- Analysis: 105 records
- Reconnaissance: 15 records
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The dominant category within the sample was Fuzzers activity. This type of traffic is commonly associated with vulnerability discovery attempts where attackers send malformed or unexpected inputs to test system behaviour and identify weaknesses.

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Most records were associated with unidentified services, while a smaller number involved FTP and HTTP services. This suggests that much of the malicious traffic did not map cleanly to common application-layer services.

The connection state distribution showed a high number of INT states compared with FIN states. This indicates that many connections were interrupted or not completed normally, which may reflect unstable or intentionally abnormal communication attempts during malicious activity.

Average packet and byte statistics also showed higher source-side activity compared with destination responses, which is consistent with aggressive outbound probing or attack-oriented traffic behaviour.

Appendix C — AI-Generated Forensic Report

The following excerpt shows the AI-generated forensic triage report produced by the locally hosted Mistral 7B model through the automated Python forensic analyzer.

=== AI AUTOMATED NETWORK FORENSICS REPORT ===

Generated: 2026-05-07 20:44:36

=== EXTRACTED TRAFFIC EVIDENCE ===

Total suspicious records analysed: 1000

Attack category distribution:

attack_cat

Fuzzers 635

Backdoor 242

Analysis 105

Reconnaissance 15

Shellcode 3

Protocol distribution:

proto

tcp 463

unas 220

udp 194

ospf 13

sctp 7

any 3

ipv6 3

rsvp 2

ib 2

ddp 1

ipv6-frag 1

cftp 1

wsn 1

pvp 1

wb-expak 1

mtp 1

pri-enc 1

sat-mon 1

cphb 1

sun-nd 1

iso-ip 1

xtp 1

il	1
mfe-nsp	1
3pc	1
ipv6-route	1
idrp	1
bn	1
swipe	1
kryptolan	1
cpnx	1
wb-mon	1
vmtp	1
dgp	1
eigrp	1
ax.25	1
gmtp	1
pnni	1
sep	1
pgm	1
idpr-cmtp	1
zero	1
rxd	1
mobile	1
narp	1
fc	1
pipe	1
ipcomp	1
ipv6-no	1
sat-expak	1
ipv6-opts	1
snp	1
ipcv	1
br-sat-mon	1
ttp	1
tcf	1
nsfnet-igp	1
sprite-rpc	1
aes-sp3-d	1
sccopmce	1
qnx	1
scps	1
etherip	1
aris	1
pim	1
compaq-peer	1

vrrp	1
iatp	1
stp	1
l2tp	1
srp	1
sm	1
isis	1
smp	1
fire	1
ptp	1
crtf	1
sps	1
merit-inp	1
idpr	1
skip	1
larf	1
ipip	1
micp	1
encap	1
ifmp	1
tp++	1
a/n	1
i-nlsp	1
ipx-n-ip	1
sdrp	1
tlsp	1
gre	1
mhrp	1
ddx	1
ippc	1
visa	1
secure-vmtp	1
uti	1
vines	1
crudp	1
iplt	1

Service distribution:

service	
-	937
ftp	38
http	22
dns	3

Connection state distribution:

state

INT 531

FIN 463

CON 6

Average source packets: 11.42

Average destination packets: 4.46

Average source bytes: 5914.77

Average destination bytes: 297.91

=== AI FORENSIC ANALYSIS ===

1. Summary of suspicious findings: The network traffic analysis shows a significant presence of Fuzzers (635 instances) and Backdoors (242 instances), which are known attack categories in the UNSW-NB15 intrusion dataset. Other suspicious activities include Reconnaissance, Shellcode, Analysis, and various protocols that are not commonly used, such as kryptol, kryptolan, mfe-nsp, pvp, wb-expak, and br-sat-mon. The service distribution shows a large number of connections to an unidentified service (-937 instances), with a smaller number of connections to FTP (38 instances) and HTTP (22 instances).

2. Likely attack behaviour present: The high occurrence of Fuzzers indicates a potential Denial-of-Service (DoS) or reconnaissance attack, as Fuzzers are used to send malformed packets to a system in order to crash it or gather information about its vulnerabilities. Backdoors suggest the presence of persistent remote access, allowing an attacker to control compromised systems undetected. Reconnaissance activities may be employed before launching more targeted attacks.

3. Severity assessment: Given the large number of Fuzzers and Backdoor instances, the severity level of this network traffic is high. The attack could potentially lead to a successful intrusion, data theft, or system crashes causing service disruptions.

4. Why this traffic would require forensic attention: Analyzing this traffic is essential to determine if any systems have been compromised, identify the attacker's tactics, and prevent further attacks. The presence of Fuzzers, Backdoors, and uncommon protocols could indicate that an advanced persistent threat (APT) is active on the network.

5. Recommended investigative actions:

- Identify systems affected by the Fuzzers and Backdoor traffic to deter

determine the extent of the compromise.

- Review log files from affected systems for indicators of compromise (IoCs), such as malicious IP addresses, unusual network activity patterns, or modified system configurations.
- Investigate uncommon protocols and services to identify potential command-and-control (C2) channels or data exfiltration routes.
- Update security controls and firewall rules to block known attack vectors and prevent further intrusion attempts.
- Implement network monitoring tools to detect similar traffic patterns in the future.
- Conduct post-incident response actions, such as patching identified vulnerabilities, cleaning affected systems, and notifying affected parties.

Project Repository

GitHub Repository:

Add GitHub repository link here.